

SPRINT 2

BLADE LOOP

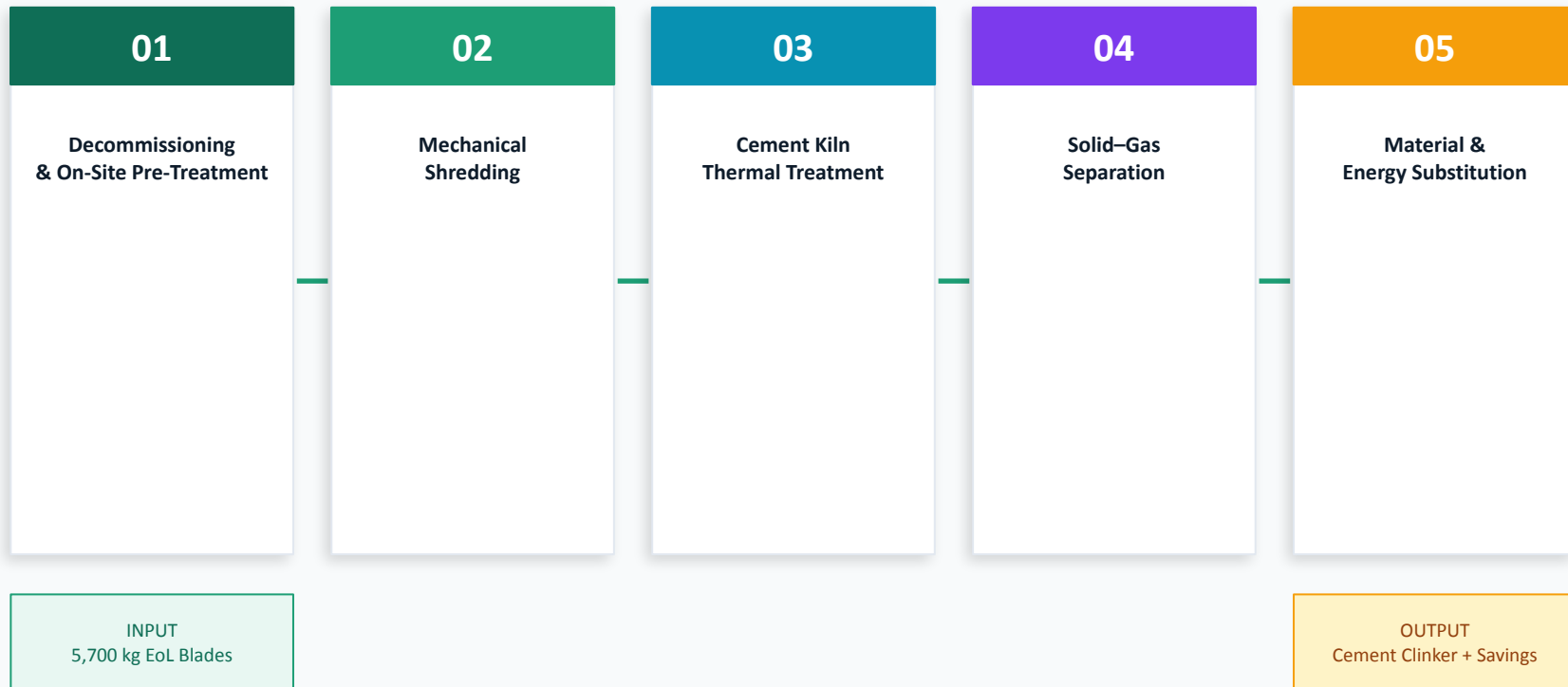
Life Cycle Intelligence for Windmill Blade Recycling

End-of-Life Process — Stage by Stage

GROUP 4 · OVGU Magdeburg · May 2026

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Process Overview — Recycling Pipeline



System boundary: blade arrival at wind farm → clinker production · Full mass & energy balance modelled at every stage

Stage 01 & 02 — Decommissioning and Mechanical Shredding

STAGE 01

Decommissioning & On-Site Pre-Treatment



Input: 5,700 kg intact EoL blades + 969 MJ diesel energy

Output: 5,700 kg blade segments (1–2 m) ready for transport

- No material loss at this stage
- Diamond wire saws used for precision root cuts
- On-site cutting reduces transport CO₂

STAGE 02

Mechanical Shredding



Input: 5,700 kg blade segments + 1.5 MWh electrical energy

Output: 5,415 kg GFRP granules (20–30 mm) + 285 kg recovered metals

- Cross-belt magnets separate all ferrous metals
- Particle size 20–30 mm critical for kiln injection
- 285 kg metal recovery = LCA recycling credit

Stage 03, 04 & 05 — Cement Kiln and Separator Stages



Input: 5,415 kg GFRP granules injected into the calciner

Output: ~4,415 kg solid mineral phase + ~1,000 kg flue gases

- Resin burns at high temperature — heat re-absorbed by kiln
- Glass minerals dissolve into the cement clinker matrix
- Clinkering zone reaches 1,450 °C



Stage 04

Solid–Gas Separation: filters flue gas, routes clean mineral oxides into cement raw meal

Stage 05

Material Substitution: blade minerals replace virgin sand & bauxite in clinker synthesis

Final Savings

342 kg

Coal Saved

177 kg

Sand Avoided

32 kg

Bauxite Saved

Mass Balance Summary — Full Process (per 5,700 kg batch)

Stage	Input (kg)	Output — Product (kg)	Output — Waste / Gas (kg)	Notes
01 — Decommissioning	5,700	5,700 (blade segments)	Minor diesel exhaust	No material loss
02 — Mech. Shredding	5,700	5,415 (GFRP granules)	285 (metals — recycled)	Metals recovered
03 — Cement Kiln	5,415	4,415 (mineral/char solid)	1,000 (flue gas)	Resin combustion heat released
04 — Solid–Gas Sep.	5,415	4,415 (clean mineral oxides)	1,000 (filtered gas)	Gas treated to EU standards
05 — Substitution	4,415	Clinker (blade minerals fused)	—	342 kg coal + 209 kg minerals saved

Conservation of mass: total inputs = total outputs (product + waste) at every stage

What This Process Means for the Dashboard

Every stage above becomes a live, interactive element in the WINDECEE Unity dashboard

5 stages → 5 animated 3D scenes

Each process stage visualised in Unity using Blender-modelled assets — shredders, kilns, separators

Mass balance → live KPI cards

The 5,700 kg input and all output quantities feed directly into the real-time dashboard KPIs

Energy & coal savings → CO₂ translator

The 342 kg coal credit powers the CO₂ savings display with flight and household equivalents

Country selector changes transport CO₂

Stage 1 transport distances differ by country — Germany, Netherlands, Denmark each produce different totals

Thank You

We welcome your questions and feedback

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